

# DS203: Programming for Data Science

Python introduction

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# Python, **python**, Python

- Language
- Interpreter
  - CPython: **python**
  - Jython: Python on the JVM
- Distribution
  - Official CPython, ActiveState Python
  - **miniforge**, Python(x,y) etc.
  - Anaconda, miniconda

# Python interpreter

- Interactive usage

```
$ python
```

```
Python 3.11.11 | packaged by conda-forge | ... [GCC 13.3.0] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>>
```

- Executing Python code (Python scripts)

```
$ python hello.py
```

```
Hello world!
```

```
$
```

# Python standard library

- Builtin additional functionality
  - Through Python modules
- “Batteries included”

# Other community packages

- Many other open source packages
- We will use many of them
- `jupyter`, `numpy`, `scipy`
- `matplotlib`, `sklearn`, `pytorch`, `tensorflow`, etc.
- Community maintained
- Often provided in a “distribution”: conda-forge etc.

# Summary: Python ecosystem

- Python Language
- Python interpreters
  - CPython: `python`
  - Jython: Python on the JVM
- Python standard library
- Community packages
- Python distributions

# Programming

## A famous quote<sup>1</sup>

Programs must be written for people to read, and only incidentally for machines to execute.

# A mental model

- `python` interprets each statement
- Top to bottom
- Sequentially



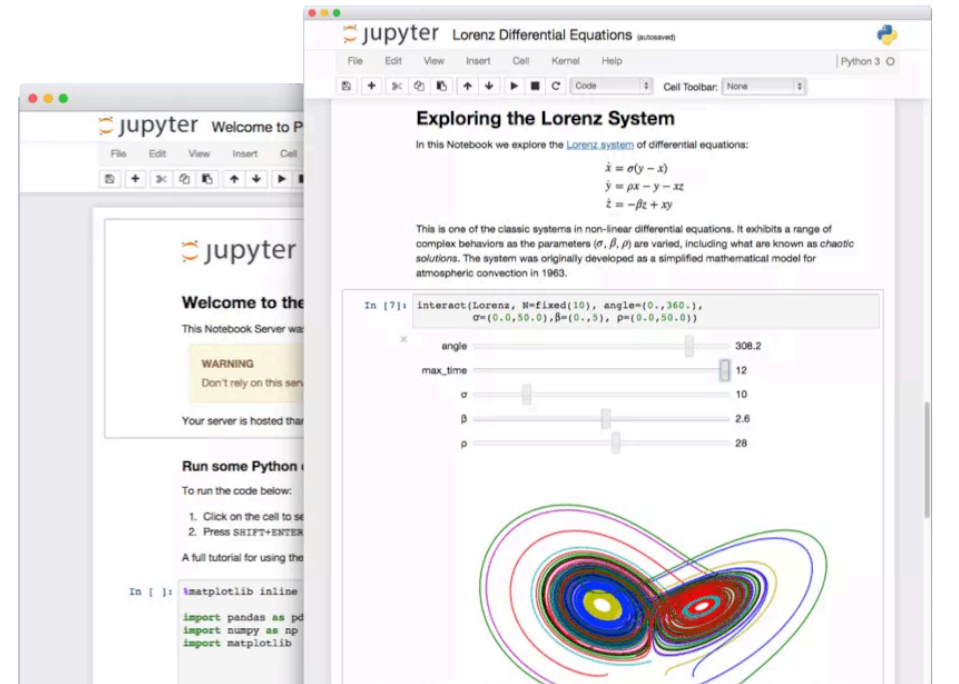
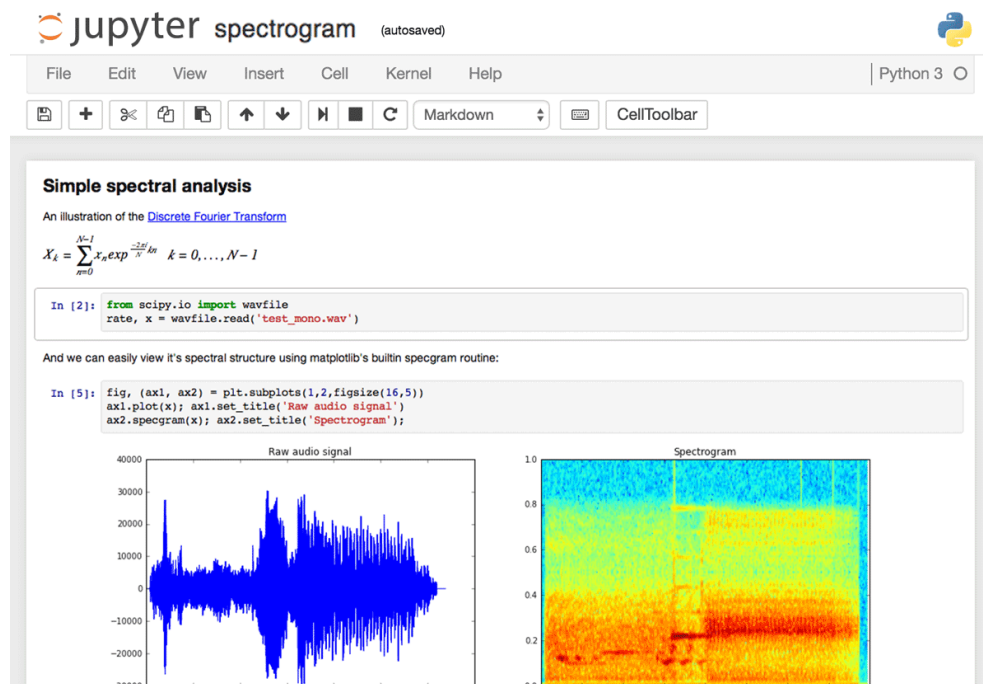
Tip

If you find programming hard, perhaps you need to provide simpler instructions?

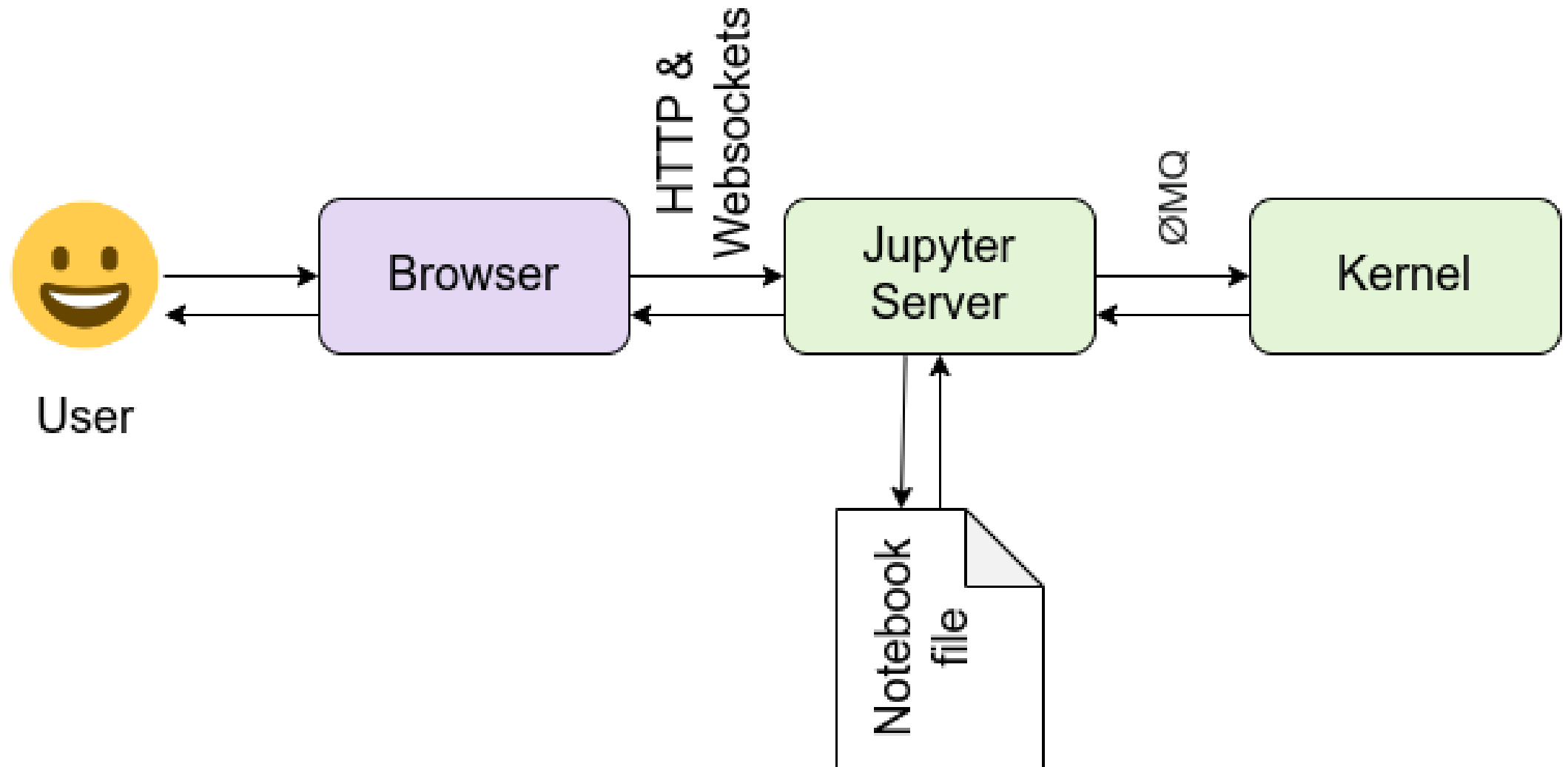


# What is Jupyter?

- Powerful interface
- Combine code, text, math
- Interactive widgets



# How does it work?



# How does it work?

- Kernels manage execution and communication
  - Supports 40 different languages
  - Python, Julia, R, Scala, Java, Octave, Scheme, Cling ...
- 
- Notebook is an interface that talks to the kernel

# Alternatives

- Jupyterlite: on your browser
- Google colab: on their servers
- Alternatives: [Marimo](#)



# Questions?

# Let us dive in!

- Visit here: [prabhuramachandran.github.io/ds203](https://prabhuramachandran.github.io/ds203)
- Provides a live environment
  - Tested on recent Firefox or Chrome
  - Don't use a private window
- Will have the class slides/notebooks etc.
- Avoid editing files in [slides](#), [tutorials](#)